Type MS · Calculate breathing rate or air volume from trace or table

Answer 1 of 4 · Question 2 part b.iii · 3 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

Number					
2	(a)	(i)	nose/nasal cavity/mouth, trachea/windpipe, bronchi, bronchioles;;	one mark for two correct two marks for four correct	2
		(ii)	• large surface area; • thin epithelium/walls;	R cell wall/cell membrane	2
	(b)	(i)	B; 0.04% A, C and D are all incorrect figures		1
		(ii)	• 1200×16; • $\frac{19200}{1000}$ = 19 dm^3;	ECF if any value used from table Accept 19.2	2
		(iii)	• $\frac{20 \times 10\,500;}{100}$ • $\frac{2100 \times 20;}{100}$ • = 420 cm^3;	2 mark ECF	3
		(iv)	• more carbon dioxide in blood/lungs/body; • detected by (chemo)receptors (in arteries); • rate of breathing/tidal volume increases; • increases to remove carbon dioxide;		4
					Total 14 marks

Type MS · Calculate breathing rate or air volume from trace or table

Answer 1 of 4 · Question 2 part b.iii · 3 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate breathing rate or air volume from trace or table

Answer 2 of 4 · Question 2 part b.i · 2 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

number				
2	(a)	• oxygen; • glucose; • water; • carbon dioxide;	in any order in any order	2 2
	(b) (i)	• A to B = 60 seconds; • 10 breaths during that period = 10 per minute;	Allow 60 anywhere in response	2
	(ii)	• 19.5 - 13.5 squares of movement = 6 squares; • $1dm^3$ = 4 squares; • 6/4 = $1.5dm^3$;		3
	(c)	• peaks/troughs higher/lower; • closer together;	Allow increase in amplitude Allow increase in frequency Answers must relate to peaks/waves/curves Ignore details relating to scientific details such as number or depth of breaths.	2

Total 11 marks

Type MS · Calculate breathing rate or air volume from trace or table

Answer 2 of 4 · Question 2 part b.i · 2 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate breathing rate or air volume from trace or table

Answer 3 of 4 · Question 2 part b.ii · 3 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

number				
2	(a)	• oxygen; • glucose; • water; • carbon dioxide;	in any order in any order	2 2
	(b) (i)	• A to B = 60 seconds; • 10 breaths during that period = 10 per minute;	Allow 60 anywhere in response	2
	(ii)	• 19.5 - 13.5 squares of movement = 6 squares; • $1dm^3$ = 4 squares; • 6/4 = $1.5dm^3$;		3
	(c)	• peaks/troughs higher/lower; • closer together;	Allow increase in amplitude Allow increase in frequency Answers must relate to peaks/waves/curves Ignore details relating to scientific details such as number or depth of breaths.	2

Total 11 marks

Type MS · Calculate breathing rate or air volume from trace or table

Answer 3 of 4 · Question 2 part b.ii · 3 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate breathing rate or air volume from trace or table

Answer 4 of 4 · Question 2 part b.ii · 2 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

Number					
2	(a)	(i)	nose/nasal cavity/mouth, trachea/windpipe, bronchi, bronchioles;;	one mark for two correct two marks for four correct	2
		(ii)	• large surface area; • thin epithelium/walls;	R cell wall/cell membrane	2
	(b)	(i)	B; 0.04% A, C and D are all incorrect figures		1
		(ii)	• 1200 × 16; • $\frac{19200}{1000}$ = 19 dm^3;	ECF if any value used from table Accept 19.2	2
		(iii)	• $\frac{20 \times 10\,500;}{100}$ • $\frac{2100 \times 20;}{100}$ • = 420 cm^3;	2 mark ECF	3
		(iv)	• more carbon dioxide in blood/lungs/body; • detected by (chemo)receptors (in arteries); • rate of breathing/tidal volume increases; • increases to remove carbon dioxide;		4

Total 14 marks

Type MS · Calculate breathing rate or air volume from trace or table

Answer 4 of 4 · Question 2 part b.ii · 2 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks